SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – TECHNICAL PRESENTATION

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 3 – Technical Presentation
Weightage:	10% of CIA	**Total Marks:**	100
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
Student Name:	A. DEKKSHITHA	**Reg. No.:**	192572133
Section / Batch:		**Date:**	28.07.2026

Instructions to Students

- Answer the given question through a technical presentation. Total Marks: 100.
- Clearly define the problem and explain the concept of algorithm growth functions.
- Present comparative analysis using appropriate representations such as graphs, tables, or charts.
- Use suitable tools (PowerPoint, visualization tools, or software) to support your explanation.
- Ensure technical accuracy, logical flow, and proper interpretation of results.
- Maintain clarity in communication, proper slide organization, and professional delivery.
- Time management, confidence, and audience engagement will be considered during evaluation.

Question Type	Focus Area	Questions
Technical Presentation	Analyze and compare growth functions using mathematical techniques and asymptotic concepts and with graphical and visual interpretation	Q1

SECTION A – Technical Presentation

Q1. Divide and Conquer Recurrence Evaluation

Analyze the recurrence T(n) = 3T(n/2) + n2 using the Master Theorem. Clearly define subproblem structure and recurrence relation. Identify parameters and justify the selected case. Use diagrams or tables for explanation. Interpret the result and discuss efficiency.

Code of Conduct

I ___________________________(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __________________________

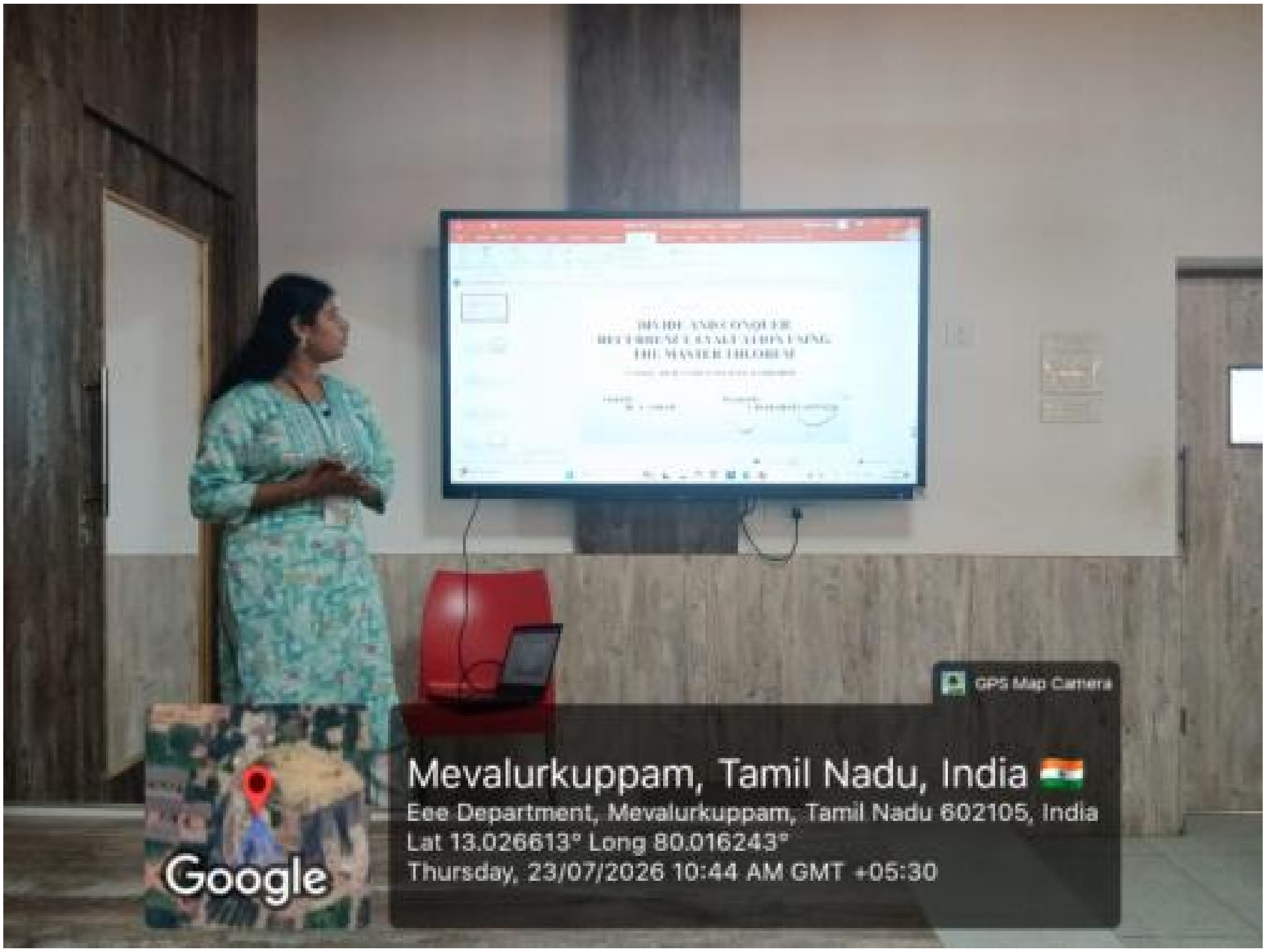
RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30%	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25%	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20%	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15%	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10%	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30%				
Problem Identification	25%				
Use of Tools	20%				
Communication	15%				
Professionalism	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________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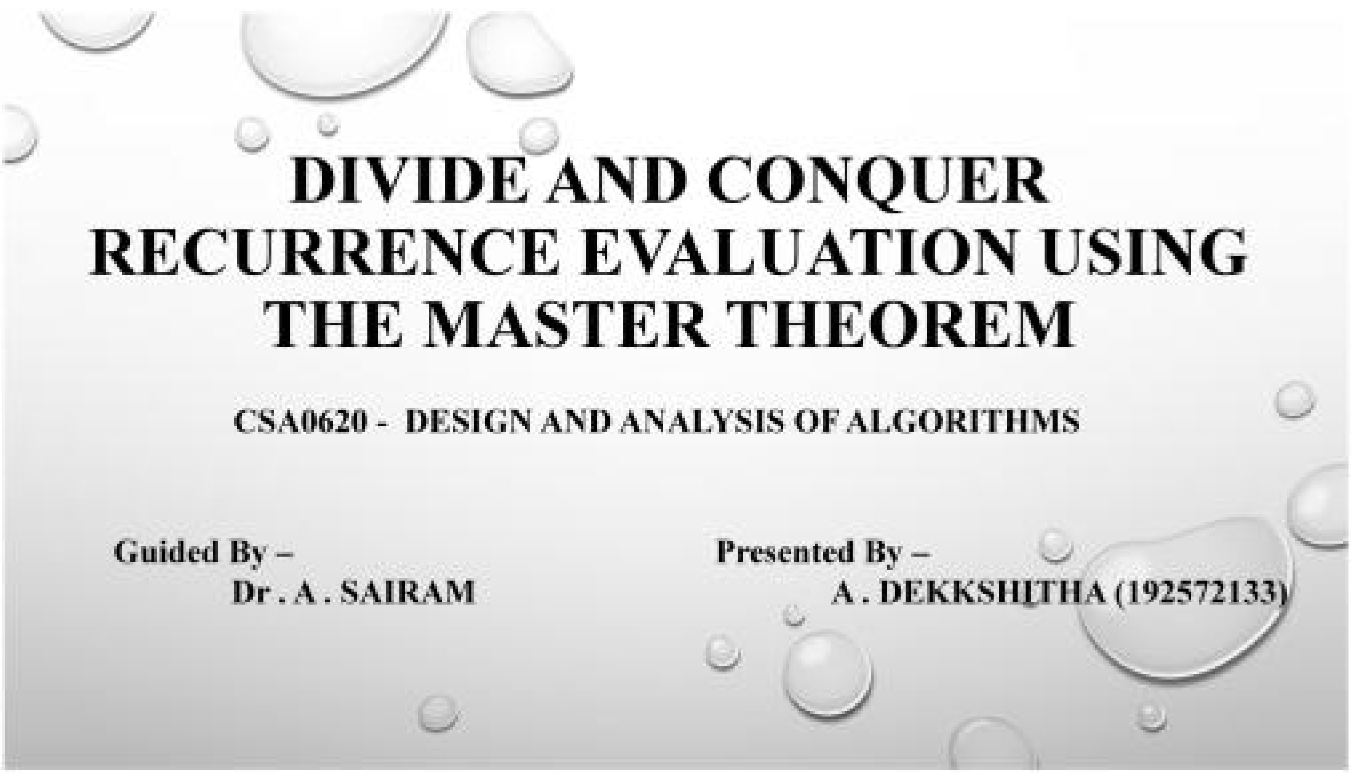

DIVIDE AND CONQUER
RECURRENCE EVALUATION USING
THE MASTER THEOREM
CSA0620 - DESIGN AND ANALYSIS OF ALGORITHMS
Guided By –
Dr . A . SAIRAM
Presented By –
A . DEKKSHITHA (192572133)

Problem Statement

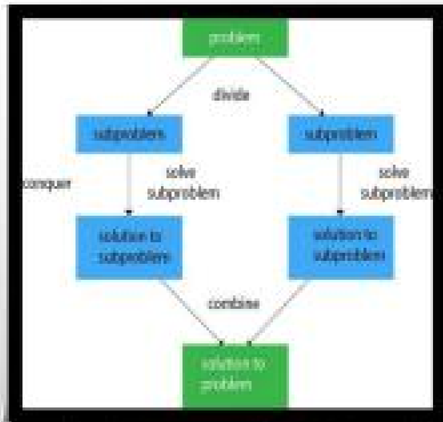
Many Divide and Conquer algorithms are analyzed using **recurrence relations**, which describe the running time of an algorithm in terms of smaller subproblems.

The given recurrence is:

$$T(n) = 3T\left(\frac{n}{2}\right) + n^2$$

Where:

- T(n) → Total running time
- 3 → Number of recursive subproblems
- n/2 → Size of each subproblem
- n^2 → Cost of dividing and combining the solutions



Algorithm Overview

The Divide and Conquer technique consists of three major steps :

- **Step 1 – Divide :**
  Split the original problem into **3 equal subproblems**, each of size **n/2**.

- **Step 2 – Conquer**
  Solve each subproblem recursively until reaching the base case.

- **Step 3 – Combine**
  Merge the solutions of all subproblems.
  The combine operation requires " n^2 " time.

Recurrence Representation :

$$T(n) = 3T(n/2) + n^2$$

Pseudocode

Pseudocode :

```
Function Solve(n)
If n == 1
    return Constant
Solve(n/2)
Solve(n/2)
Solve(n/2)
Perform n² combine operation
Return Result
```

Recurrence Relation :

$$T(n) = 3T(n/2) + n^2$$

Base Case :

$$T(1) = \Theta(1)$$

Working Example (Dry Run)

Assume ,

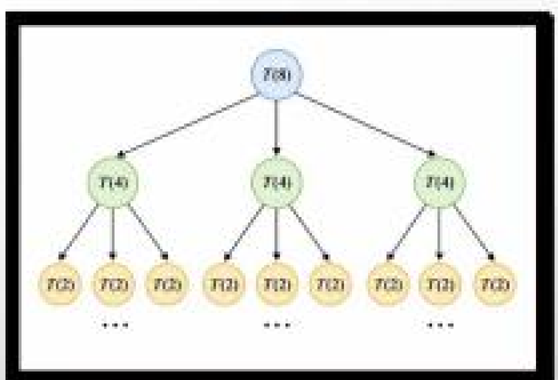
$$n = 8$$

Then ,

$$\begin{aligned} T(8) &= 3T(4) + 64 \\ &= 9T(2) + 3(16) + 64 \\ &= 27T(1) + 9(4) + 48 + 64 \end{aligned}$$

Base case is reached.
The total running time is obtained by adding the recursive work and combine work at every level.

Recursion Breakdown :



Recursion Tree Analysis

- **Recursion Tree**
- At every recursive level:

Level	Number of Subproblems	Cost per Subproblem	Total Cost
0	1	n^2	n^2
1	3	$(n/2)^2$	$\frac{3n^2}{4}$
2	9	$(n/4)^2$	$\frac{9n^2}{16}$
3	27	$(n/8)^2$	$\frac{27n^2}{64}$
k	3^k	$(n/2^k)^2$	$n^2\left(\frac{3}{4}\right)^k$

Total Work

$$T(n) = n^2 + n^2\left(\frac{3}{4}\right) + n^2\left(\frac{3}{4}\right)^2 + \cdots$$

This forms a **geometric series** with common ratio:

$$\frac{3}{4} < 1$$

Hence,

$$T(n) = \Theta(n^2)$$

Observation:

- The work performed decreases at each level.
- Therefore, the total work is dominated by the first few levels.

Complexity Analysis (Master Theorem)

General form:

$$T(n) = aT(n/b) + f(n)$$

- **Step 1 – Identify Parameters**

Parameter	Value
a	3
b	2
f(n)	n^2

- **Step 2 – Calculate**

$$n^{\log_b a} = n^{\log_2 3}$$

Since,

$$\log_2 3 \approx 1.585$$

Therefore,

$$n^{1.585}$$

- **Step 3 – Compare**

$$n^2 > n^{1.585}$$

Therefore,

$$f(n) = \Omega(n^{1.585+\epsilon})$$

- **Step 4 – Regularity Condition**

$$3f(n/2) = 3\left(\frac{n}{2}\right)^2 = \frac{3n^2}{4}$$

Since ,

$$\frac{3}{4} < 1$$

the regularity condition is satisfied.

Final Result :

- ❑ This satisfies **Master Theorem Case 3**.
- ❑ Therefore,

$$\boxed{T(n) = \Theta(n^2)}$$

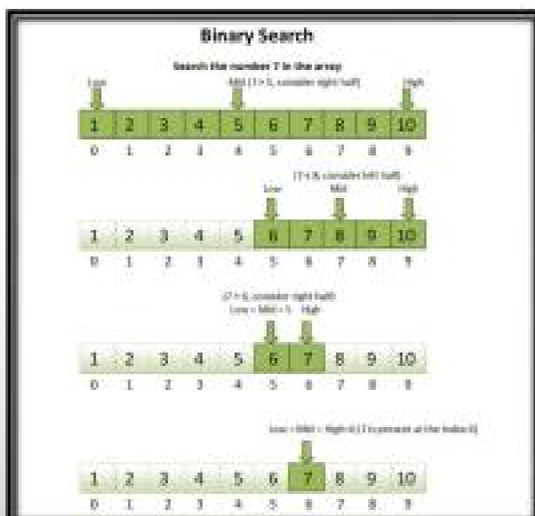
Applications

Applications :

- Merge Sort
- Quick Sort
- Binary Search
- Strassen Matrix Multiplication
- Karatsuba Multiplication
- Closest Pair of Points
- Fast Fourier Transform (FFT)

Benefits :

- Efficient recursive computation
- Reduced problem size
- Easier complexity analysis
- Scalable algorithm design



Results & Discussion

Results :

The recurrence

$$T(n) = 3T(n/2) + n^2$$

was analyzed using the Master Theorem.

Findings :

- Number of recursive calls = **3**
- Subproblem size = **n/2**
- Combine cost = **n²**
- Applicable Master Theorem Case = **Case 3**
- Time Complexity =

$$\boxed{\Theta(n^2)}$$

Discussion :

- The combine operation grows faster than the recursive work.
- Recursive cost decreases at every level.
- The recursion tree confirms the Master Theorem result.
- Therefore, the algorithm has **quadratic time complexity**.

Conclusion

- Recurrence relation successfully analyzed.
- Divide and Conquer structure identified.
- Parameters **a**, **b**, and **f(n)** determined.
- Master Theorem Case 3 applied correctly.
- Regularity condition verified.
- Final complexity obtained as

$$\boxed{\Theta(n^2)}$$

The Master Theorem provides a simple and efficient way to analyze the time complexity of Divide and Conquer algorithms.

References

• Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest & Clifford Stein, **Introduction to Algorithms (CLRS)**, 4th Edition, MIT Press.

• Ellis Horowitz, Sartaj Sahni & Sanguthevar Rajasekaran, **Fundamentals of Computer Algorithms**, Universities Press.

•Alfred V. Aho, John E. Hopcroft & Jeffrey D. Ullman, **The Design and Analysis of Computer Algorithms,** Pearson.